Proxmox VE Cluster - Zero-Downtime Rolling Node Renaming SOP

Complete Technical Procedure, Architecture Strategy & Execution Command Runbook

CLUSTER	NODE MAPPING	WORKLOAD IMPACT	CEPH & COROSYNC STATUS
cisai-pve (3-Node High Availability)	prox22, 23, 24 ➔ prox022, 023, 024	0 seconds VM downtime (Live Migrated)	HEALTH_OK • Quorate (3/3 votes)

Execution Summary: Officially, Proxmox VE does not support in-place node renaming for clustered nodes due to hardcoded dependencies across Corosync, Ceph CRUSH hierarchies, pmxcfs cluster directories, and SSL certificates. This document outlines the rolling orchestration methodology used to safely rename all three physical hypervisors from `prox22, prox23, prox24` to `prox022, prox023, prox024` while maintaining 100% guest uptime for live production VMs.

1. Cluster Architecture & Initial Mapping

Baseline Environment

Legacy Hostname	Target Hostname	Management IP	Initial Workloads & Storages	Ceph Daemons
prox22	**prox022**	`10.31.6.172`	VM 100, 102, 106 (250GB local-lvm), 9001, 9002, 9004	mon.prox22, mgr.prox22, 4 OSDs (hdd)
prox23	**prox023**	`10.31.6.173`	VM 9003 (zfs-ha replicated pool)	mon.prox23, mgr.prox23, 4 OSDs (hdd)
prox24	**prox024**	`10.31.6.174`	VM 103 (150GB local-lvm), VM 104 (150GB local-lvm)	mon.prox24, mgr.prox24, 4 OSDs (hdd)

2. Technical Strategy: Why Rolling Execution Works

Zero Downtime Principles

- **Dual-DNS Alias Resolution:** Updating `/etc/hosts` with both old and new names across all nodes ensures that Corosync and SSH can resolve all peers continuously during the transition window.
- **Ceph CRUSH Bucket Renaming:** Running `ceph osd crush rename-bucket <old> <new>` updates the topology map instantly without triggering any data rebalancing or I/O degradation.
- **Ceph Monitor/Manager Re-creation:** Because Ceph requires 2 of 3 monitors for quorum, destroying 1 monitor at a time on the maintenance node allows the cluster to maintain quorum on the surviving 2 nodes.
- **Corosync Version Incrementing:** Incrementing `config_version` in `/etc/pve/corosync.conf` propagates the new nodelist across pmxcfs automatically.
- **Preservation of Local LVM Volumes:** Virtual machines with disks on `local-lvm` (e.g. VM 106 with 250GB) never leave their physical host. Only the VM configuration pointer in `/etc/pve/nodes/` moves to the new folder.

3. Pre-Flight Setup: Dual-DNS Synchronization

Step 0 • All Nodes

Execute on all 3 nodes (10.31.6.172, 10.31.6.173, 10.31.6.174)

Ensures seamless resolution for both legacy and new hostnames simultaneously.

```
cat << 'EOF' > /etc/hosts
127.0.0.1 localhost.localdomain localhost
10.31.6.172 prox022 prox22
10.31.6.173 prox023 prox23
10.31.6.174 prox024 prox24

::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
EOF
```

4. Phase 1: Rolling Rename of Node 1 (prox23 ➔ prox023)

Node 1 Procedure

Step 1.1: Live-Migrate VM 9003 to prox24

Empties all workloads from prox23 before initiating maintenance.

```
ha-manager migrate vm:9003 prox24
```

Step 1.2: Destroy Ceph MON & MGR on prox23

Run directly on 10.31.6.173. Quorum remains healthy (2/3 monitors online).

```
pveceph mon destroy prox23
pveceph mgr destroy prox23
```

Step 1.3: Rename Ceph CRUSH Host Bucket

Prevents placement group (PG) reshuffling and maintains OSD health.

```
ceph osd crush rename-bucket prox23 prox023
```

Step 1.4: Place Node in HA Maintenance & Update Configs

Disables watchdog fencing and updates Corosync, Storage, and Replication.

```
ha-manager crm-command node-maintenance enable prox23
sed -i "s/prox23/prox023/g" /etc/pve/storage.cfg
sed -i "s/prox23/prox023/g" /etc/pve/replication.cfg
sed -i "s/name: prox23/name: prox023/g" /etc/pve/corosync.conf
sed -i "s/config_version: 4/config_version: 5/g" /etc/pve/corosync.conf
```

Step 1.5: Set Hostname, Restart Services & Regenerate SSL (on 10.31.6.173)

Applies new hostname and rebuilds node certificates.

```
hostnamectl set-hostname prox023
echo prox023 > /etc/hostname
systemctl restart pve-cluster corosync
pvecm updatecerts -f
systemctl restart pvedaemon pveproxy pvestatd pve-ha-lrm pve-ha-crm
```

Step 1.6: Re-establish Ceph Daemons & Disable Maintenance

Rebuilds mon and mgr daemons under the new identity and restores HA.

```
pveceph mon create
pveceph mgr create
ha-manager crm-command node-maintenance disable prox023
rm -rf /etc/pve/nodes/prox23
```

5. Phase 2: Rolling Rename of Node 2 (prox24 ➔ prox024)

Node 2 Procedure

Step 2.1: Relocate Replicated Workload Back to prox023

```
ha-manager migrate vm:9003 prox023
```

Step 2.2: Failover Ceph Active Manager & Destroy Daemons

If prox24 is active Ceph manager, trigger clean failover to standby.

```
ceph mgr fail prox24
# On 10.31.6.174:
pveceph mon destroy prox24
pveceph mgr destroy prox24
```

Step 2.3: Update Ceph CRUSH, HA Maintenance & Configs

```
ceph osd crush rename-bucket prox24 prox024
ha-manager crm-command node-maintenance enable prox24
sed -i "s/prox24/prox024/g" /etc/pve/storage.cfg
sed -i "s/prox24/prox024/g" /etc/pve/replication.cfg
sed -i "s/name: prox24/name: prox024/g" /etc/pve/corosync.conf
sed -i "s/config_version: 5/config_version: 6/g" /etc/pve/corosync.conf
```

Step 2.4: Update Hostname, Restart Services & Relocate VM Configs

VM 103 and 104 stay running on local-lvm without interruption.

```
# On 10.31.6.174:
hostnamectl set-hostname prox024
echo prox024 > /etc/hostname
systemctl restart pve-cluster corosync
pvecm updatecerts -f
systemctl restart pvedaemon pveproxy pvestatd pve-ha-lrm pve-ha-crm
mv /etc/pve/nodes/prox24/qemu-server/*.conf /etc/pve/nodes/prox024/qemu-server/
rm -rf /etc/pve/nodes/prox24
```

Step 2.5: Recreate Ceph Daemons & Restore HA

```
pveceph mon create
pveceph mgr create
ha-manager crm-command node-maintenance disable prox024
```

6. Phase 3: Rolling Rename of Final Node (prox22 ➔ prox022)

Node 3 Procedure

Step 3.1: Live-Migrate Shared Storage VMs Off prox22

Migrate Ceph-backed VMs (9001, 9002, 9004) to prox024 and prox023. Local VMs (100, 102, 106) stay on prox22.

```
ha-manager migrate vm:9001 prox024
ha-manager migrate vm:9002 prox024
ha-manager migrate vm:9004 prox023
```

Step 3.2: Ceph Failover & Daemon Removal on prox22

```
ceph mgr fail prox22
# On 10.31.6.172:
pveceph mon destroy prox22
pveceph mgr destroy prox22
ceph osd crush rename-bucket prox22 prox022
```

Step 3.3: HA Maintenance & Config Updates (Executed via Helper prox023)

```
ha-manager crm-command node-maintenance enable prox22
sed -i "s/prox22/prox022/g" /etc/pve/storage.cfg
sed -i "s/prox22/prox022/g" /etc/pve/replication.cfg
sed -i "s/name: prox22/name: prox022/g" /etc/pve/corosync.conf
sed -i "s/config_version: 6/config_version: 7/g" /etc/pve/corosync.conf
```

Step 3.4: Hostname Change & VM Pointer Synchronization

Local VMs 100, 102, and 106 (250GB) maintain uninterrupted disk access.

```
# On 10.31.6.172:
hostnamectl set-hostname prox022
echo prox022 > /etc/hostname
systemctl restart pve-cluster corosync
pvecm updatecerts -f
systemctl restart pvedaemon pveproxy pvestatd pve-ha-lrm pve-ha-crm
mv /etc/pve/nodes/prox22/qemu-server/*.conf /etc/pve/nodes/prox022/qemu-server/
rm -rf /etc/pve/nodes/prox22
```

Step 3.5: Rebuild Ceph & Re-balance Cluster

```
pveceph mon create
pveceph mgr create
ha-manager crm-command node-maintenance disable prox022
ha-manager migrate vm:9001 prox022
```

7. Post-Migration Verification & Health Validation

Verification Audit

Subsystem	Verification Command	Expected State	Audit Result
Corosync Quorum	`pvecm status`	Quorate: Yes, 3 nodes (prox022, prox023, prox024)	**PASSED (Config v7)**
Cluster Nodes	`pvecm nodes`	All 3 nodes online, nodeids 1, 2, 3 active	**PASSED (All Active)**
Ceph Health	`ceph status`	HEALTH_OK • 3 MONs in quorum • 12/12 OSDs up & in	**PASSED (Zero I/O Error)**
HA Manager	`ha-manager status`	quorum OK, active CRM master, all LRMs active	**PASSED (All Active)**
Virtual Machines	`qm list`	All 9 VMs running across the 3 nodes with intact PIDs	**PASSED (0s Downtime)**

8. Automation Reference

Script Artifact

The full automated Python script is stored locally for future disaster recovery and cluster rebuilds:

```
/home/nishusaini/Desktop/Innspark/Proxmox Cluster/Testing/cluster_rolling_rename_prox022_prox023_prox024.py
```